

RIFT Phase 2 — Simulation Summary (SIM112–SIM120)

Sim	Test	Result	Key metric
SIM112	$\lambda \Psi^4$ quintessence	FAIL	$w_a > 0$ (freezing)
SIM113	SSB hilltop $\lambda(\Psi^2 - v^2)^2$	PARTIAL	$w_0 = -0.973$, $w_a = -0.41$, 2.7σ DESI
SIM114	$\beta \Psi^2 \rho_m$ condensate	FAIL	Structural trilemma
SIM115	Gradient soliton	FAIL	$\rho_{\text{DM}} \sim 10^{-14} \times$ needed
SIM116	Ψ -switch ξ condensate	FAIL	H_0^2 suppression ²
SIM117	Level-2 recursion	FAIL	$r_\xi \sim$ Gpc (scale tension)
SIM118	RIFT-seeded fuzzy χ DM	PARTIAL	$m_{22} = 0.082$, 18 PASS, $\chi^2/\text{dof} = 0.34$
SIM119	SPARC-wide fuzzy χ DM	PASS	65/161 PASS, $m_{22}^{\text{med}} = 0.28$
SIM120	Joint DE+DM background	PASS	Decoupling exact, $\kappa = 3.8 \times 10^{-103}$
SIM120-alt	Universal m_{22} (virial)	PARTIAL	$m_{22} = 0.060$, outside window